

Optimizing SSD Performance Testing Using FIO Sprandom

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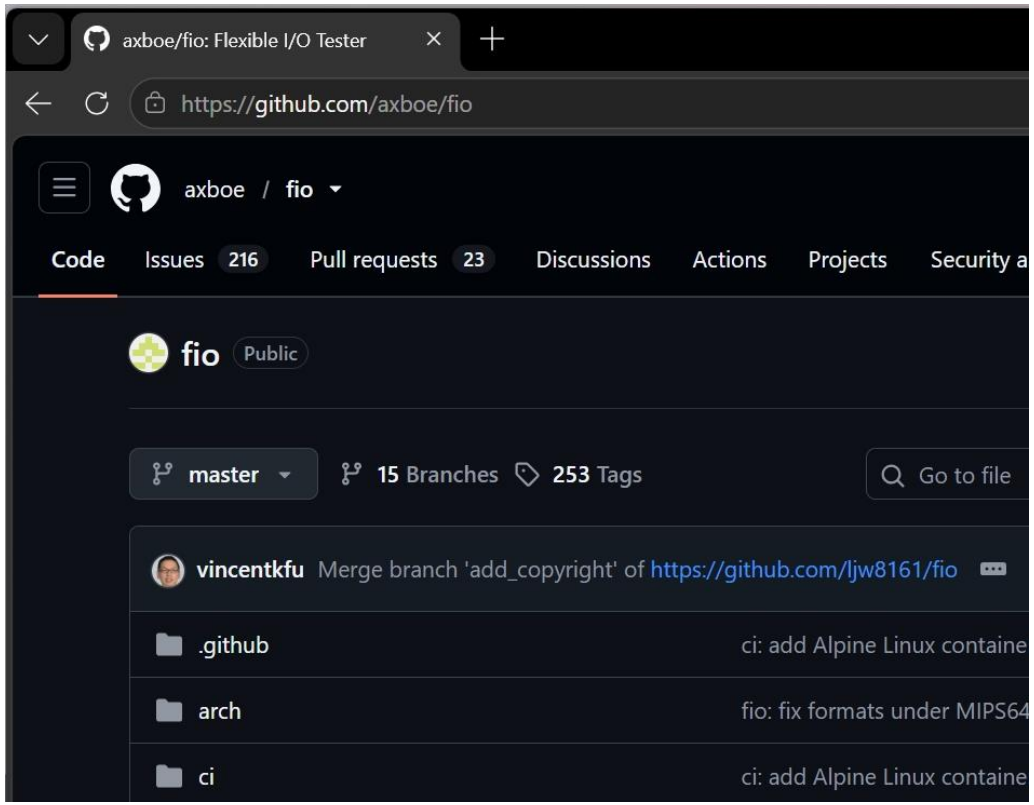


Agenda

- Sprandom Overview
- Sprandom with Protection Information (PI)
- Sprandom with Changing Host Utilization

Overview: What is Sprandom?

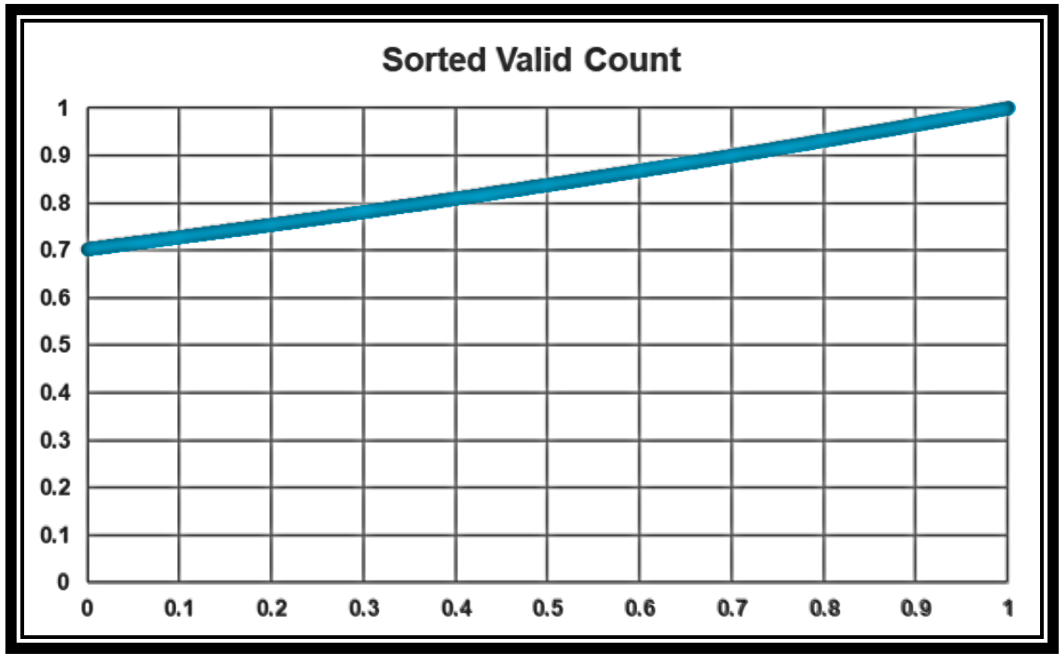
- Steady state random write performance reflects real-world performance levels
- Measured by recording the final performance levels of a random write workload for hours or even days
  - Steady State is required for OCP Performance tests
- Sprandom reduces the preconditioning time from days to hours for large devices
- Available as a feature in Fio
- Introduced by Steven Sprouse and colleagues in 2025: link



FIO

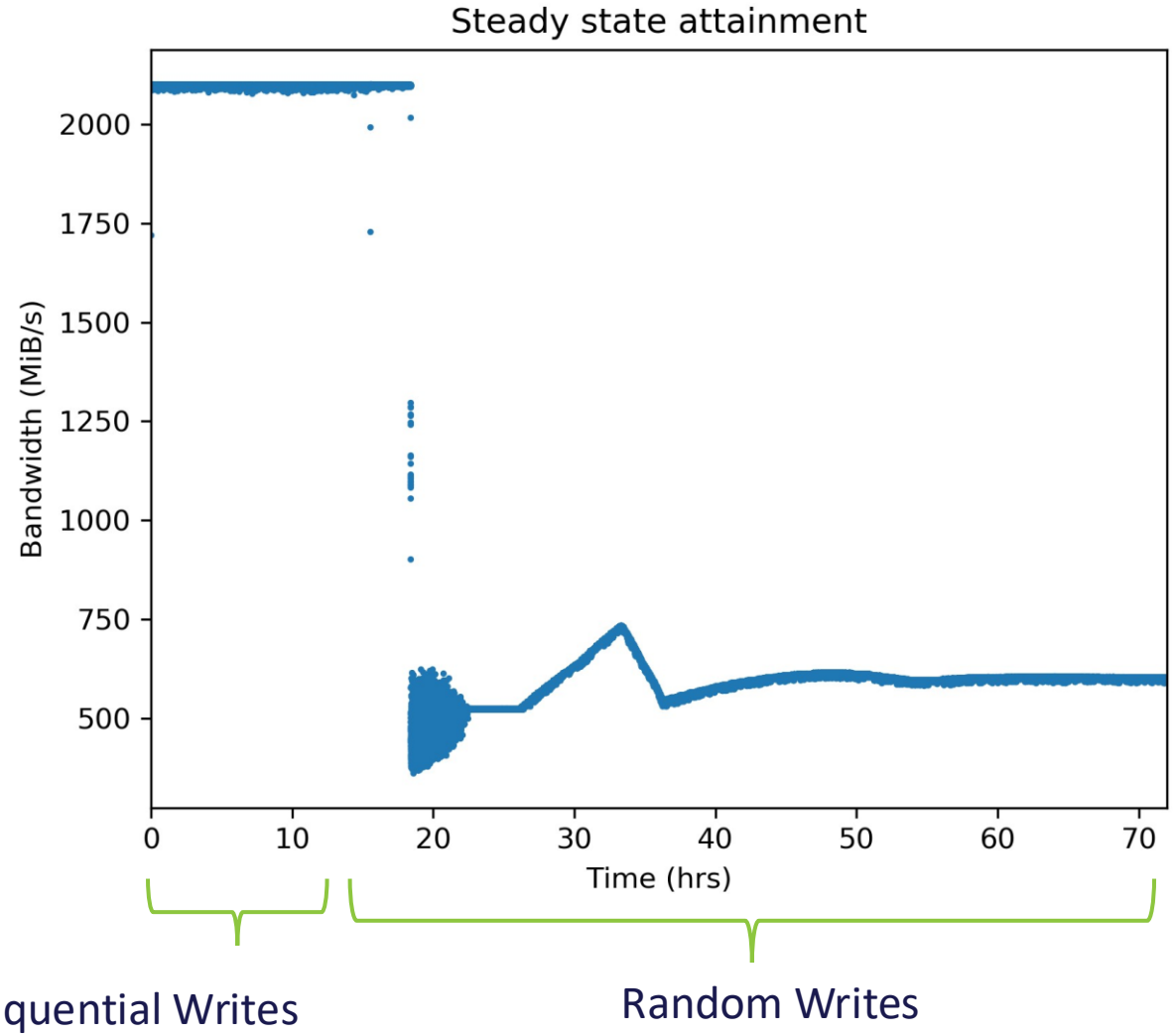
Overview: How does Sprandom work?

- With a greedy garbage collection algorithm, we can analytically derive the device's distribution of valid and invalid pages at steady state
- Some erase blocks will have more valid pages, others will have fewer valid pages
- Instead of using hours or days of random writes to achieve this state, Sprandom writes the device's physical capacity in a single pass to achieve this distribution by intentionally choosing the LBAs to write and overwrite



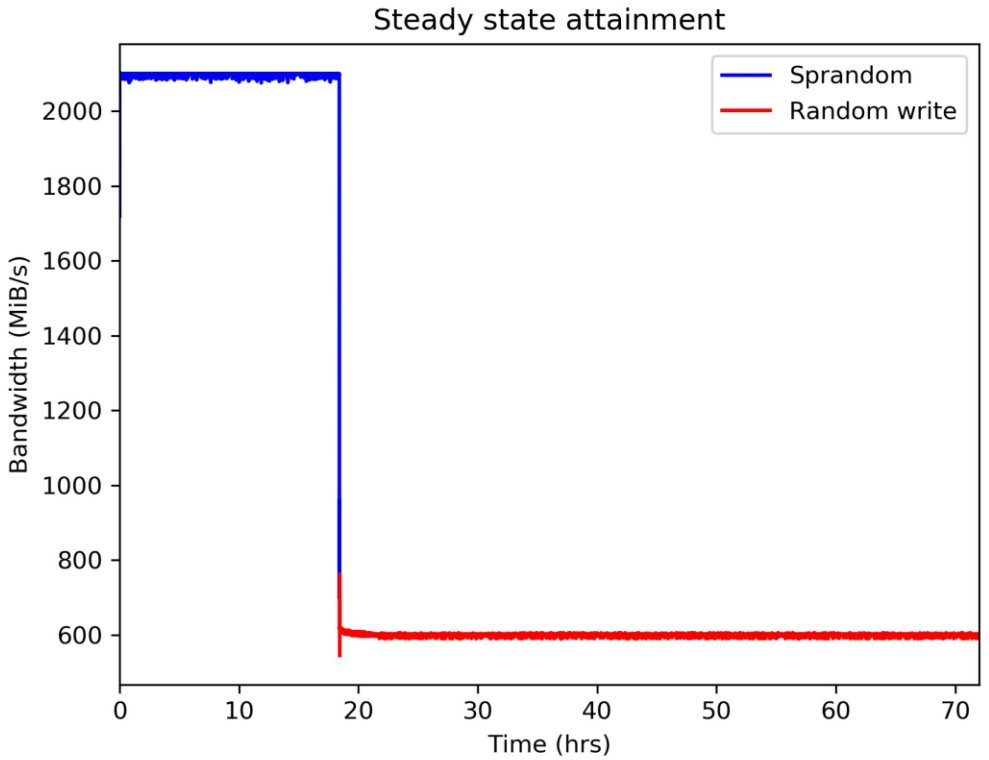
Overview: Example Conventional Run

- Using random writes to attain steady state performance levels is a long, slow process
- A standard random write workload requires 2.5 days to achieve sustained performance levels



Overview: Sprandom run

- What previously required days to achieve is now possible to accomplish in less than one day with Sprandom
- Sprandom writes at peak performance levels at 2+ GiB/s for ~18hrs
- During the subsequent random write job the the device instantly drops to sustained 600 MiB/s performance levels



Overview: Sprandom Usage Notes

- Fio blocksize parameter should be IU size to avoid read-modify-write
- Sprandom requires two parameters: device OP and number of regions
  - Outcomes are sensitive to device OP
  - Outcomes not sensitive to number of regions

```
root@localhost:~# ./fio --name=sprandom --ioengine=io_uring --iodepth=64 --bs=16k \
> --sprandom=1 --spr_op=0.200 --spr_num_regions=3000 --filename=/dev/nvme0n1 \
> --rw=randwrite --random_generator=lfsr --norandommap=1
```

- spr_cs={cache size} option delays invalidating writes to prevent original and over-write from being present in device cache at the same time
- For the *subsequent* random write workload use an updated Fio or explicitly set random_generator=tausworthe64 for larger devices

```
root@localhost:~# ./fio --name=randwrite --ioengine=io_uring --iodepth=64 --bs=16k \
> --filename=/dev/nvme0n1 --rw=randwrite --random_generator=tausworthe64
```

Extending Sprandom

- After determining appropriate Sprandom parameters, how do we extrapolate to new use-cases?

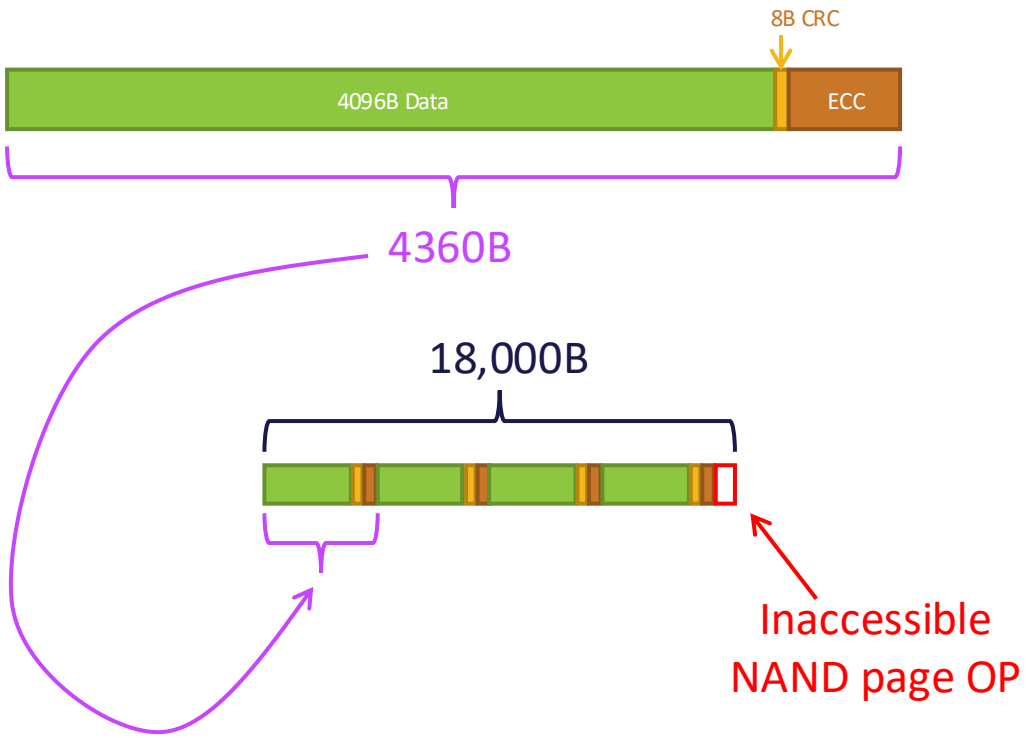
- Examples:

  - Occasionally changing Protection Information (PI) settings will change SSD logical capacity

  - Steady State performance at varying Host Utilization settings

Sprandom with Protection Information (PI)

- Example 4096B Data Flow through Controller
  - 4096B Write data transferred into Controller
  - Controller adds 8B CRC – Protect against silent data corruption
  - Optionally Controller adds Metadata (not shown)
  - Controller adds ECC – Protect against data loss
    - Ex: 256B of ECC
  - Controller writes everything to NAND
    - 4096+8+256 = 4360B
- NAND Page Sizing
  - Selected before ECC engine is designed, and must propagate to all product lines
  - Four 4096B writes are written to a NAND page
  - Extra space is likely to remain due to misalignment
    - Example NAND page size: 18,000B
    - 4360B x 4 = 17,440B
    - 17,440B does not consume all of the 18,000B
- **Takeaway:** Each NAND page has inaccessible OP
  - NAND page OP is used for CRC, ECC, and other metadata
    - Critical to drive function
  - NAND page OP is inaccessible otherwise (mostly)



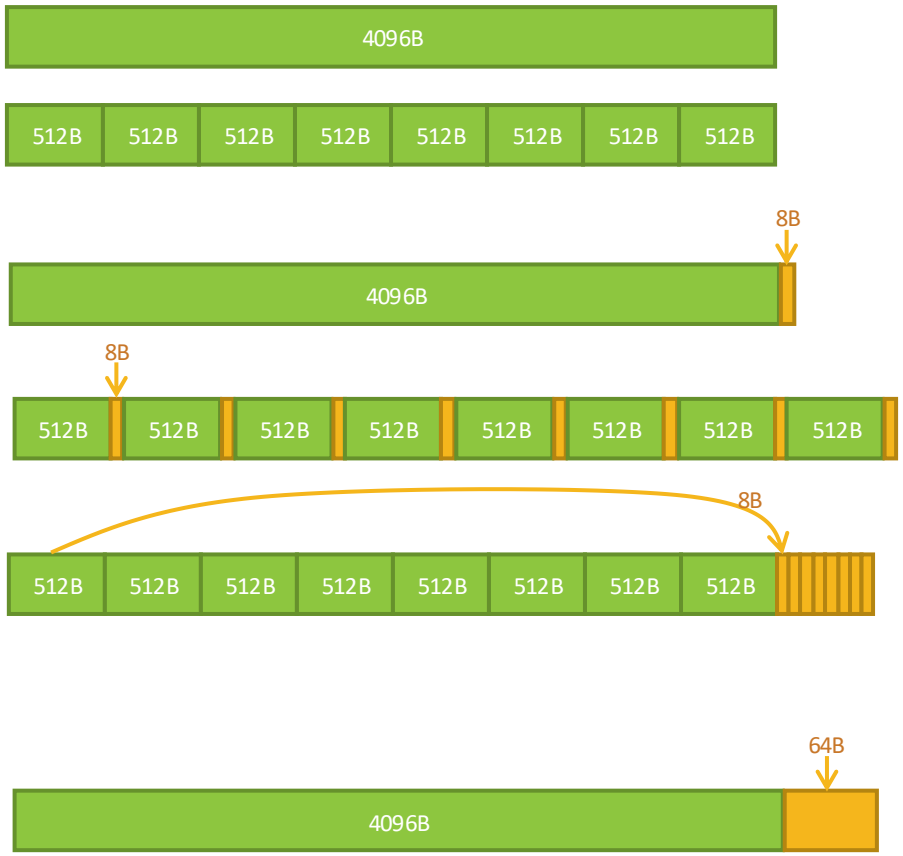
PI Options and NAND Page Sizing

- **Potential** options for PI that might fit within NAND Page Sizing

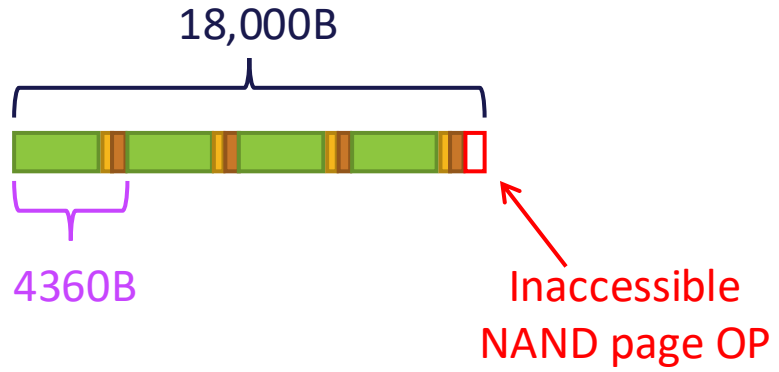
- No PI – *Internal CRC added*
- Standard “old fashioned” style PI of DIF and DIX varieties
- Some of the new “Extended LBA Formats”

NAND page sizing decisions will vary by generation and vendor

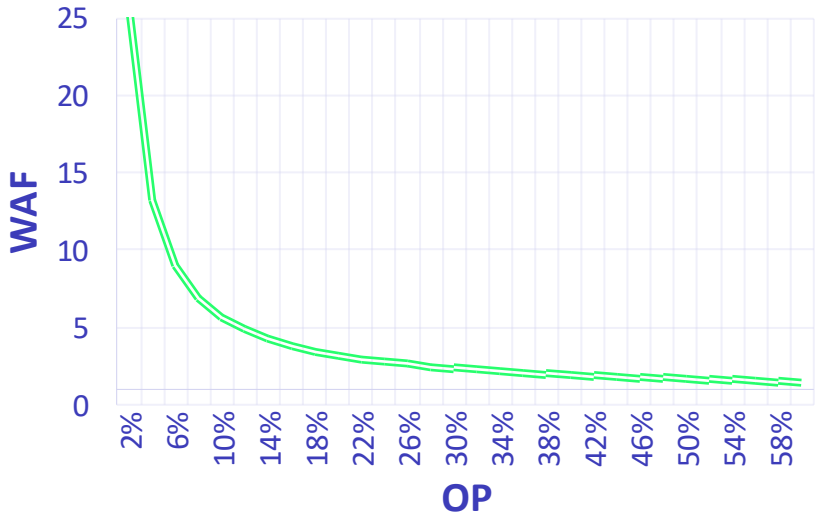
- Precise list of PI choices will require a vendor discussion



Clarifying the Definition of OP



- Does the “inaccessible OP” help WAF?
  - Usually no.
  - Only if implementing some compaction algorithms to make the OP accessible and use this area for partial sectors



- WAF due to GC
  - Depends on number of destinations for the GC data
  - Methods of using NAND page spacing as described here keeps the number of destinations for GC data constant

When Different PI Change SSD Logical Capacity

- Sometimes a change in PI settings causes a change in the logical capacity of the SSD
- *When the PI fits within the NAND page size*, the following formula may be used to change the `--spr_op` value with changing PI selections
  - Solve for the number of places on the NAND available for holding a sector

```
Phy_Sectors = LBAs_orig * (1+spr_op_orig)
```

  - Solve for the new `--spr_op` value

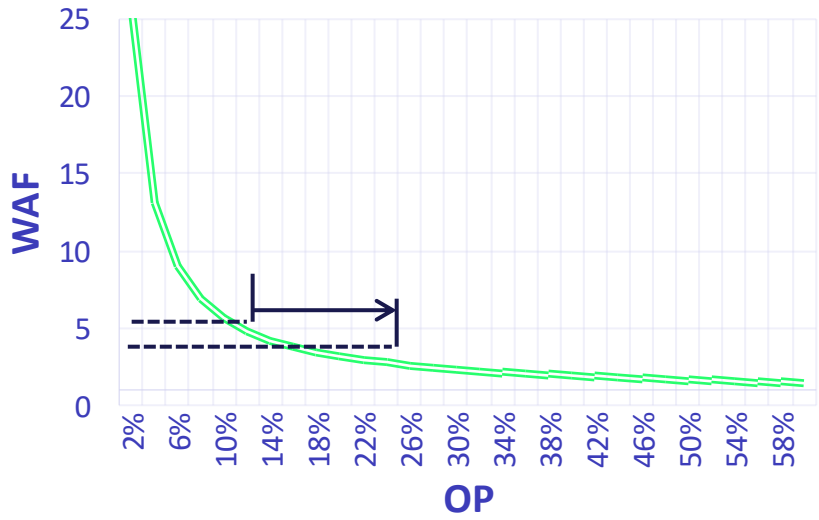
```
spr_op_new = 1 - (Phy_Sectors / LBAs_new)
```

- `LBAs_orig` and `LBAs_new` can be read from `nsze` before and after the PI format change
  - `nvme id-ns /dev/nvme0n1 -H`

- We measured approximately a 2.5% change to `--spr_op` from a PI change of 0B to 64B.
  - This OP change is because the number of LBAs reduced with the increasing PI.
  - The LBA reduction is an immediate OP improvement for the SSD

Sprandom with Changing Host Utilization

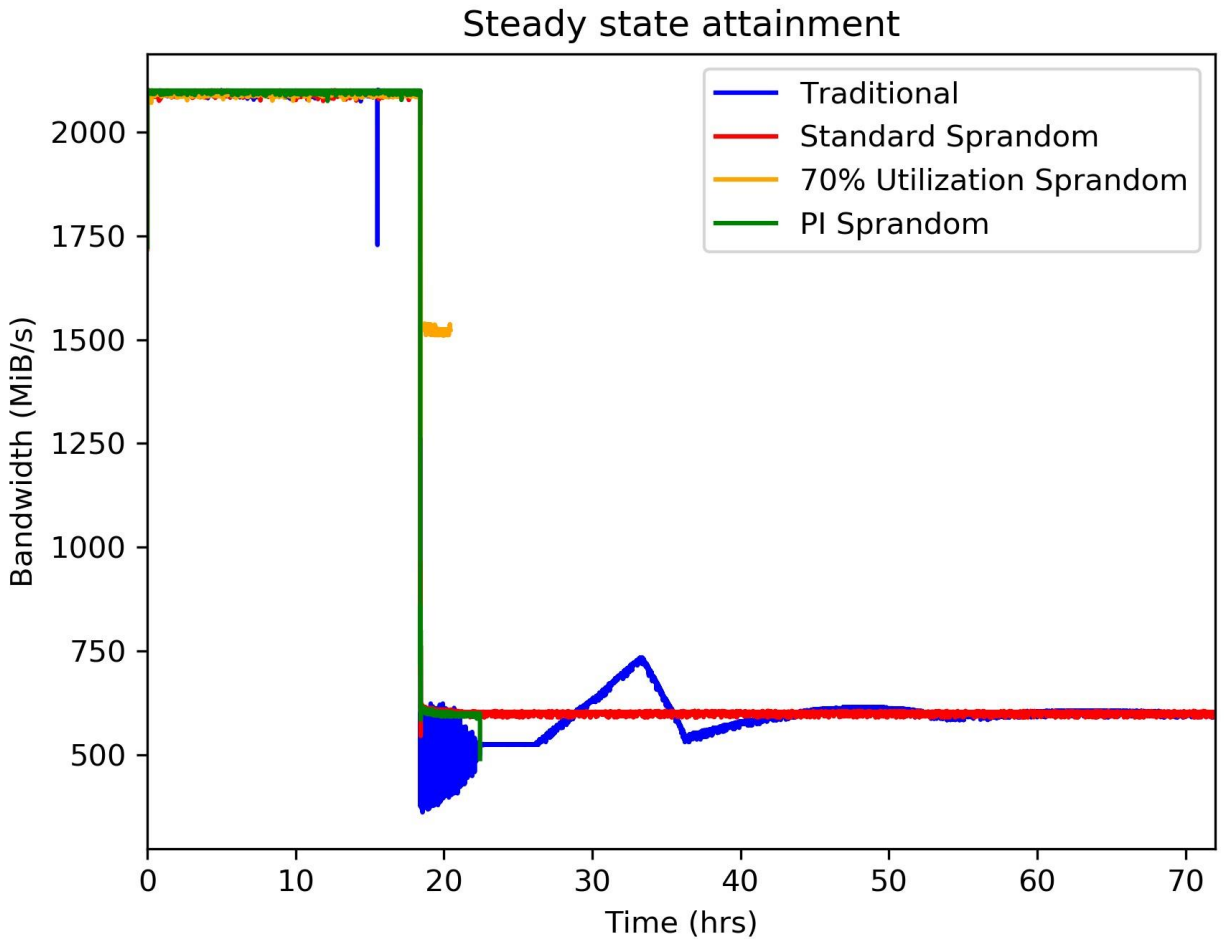
- Defining Host Utilization:
  - Utilization is a percentage of SSD logical capacity that is written by the Host.
  - Utilization may be less than 100% if the LBAs are never written or the LBAs are deallocated (TRIM/Unmapped).
- L2P Table entries that are deallocated or never written may point to a signal value (Ex: 0xFFFF) rather than consume a physical location on the NAND
  - The SSD only writes and consumes physical space for real written data.
- Reduced Utilization Improves OP by the following formula
  - `spr_op_new = 1 – ((1+spr_op_orig) / utilization)`



Results

- Run times
  - Traditional
    - 60hrs
  - Sprandom (all varieties)
    - 18hrs
    - **Recommended:** Random Writes after Sprandom to initialize any internal controller states
  - ~1/3 of the total run time

- Performance Comments
  - Traditional and Sprandom have the same Steady State performance
  - PI settings for this drive has negligible performance change
  - Reduced Utilization has significant performance improvement due to OP increases



Summary

- Introduced Sprandom and Provided Usage Guidance
  - Recommended alignment between `--bs` and IU size
  - Robust to `--spr_regions` changes
  - Sensitive to `--spr_op` changes



FIO



OCP NVMe SSD spec

- Extended Sprandom's Utility after having Initial Parameters
  - Provided the formulas for propagating PI changes into `--spr_op` changes
    - **Assuming** the PI changes fit within the same NAND page sizing
  - Provided the formula for translating a reduced host utilization into an improvement in `--spr_op`

- Sprandom can accelerate your performance qualifications for OCP NVMe SSDs
  - It's a very versatile tool that successfully accelerates Steady State Random Write Preconditioning for SSDs.

Thank You!

